



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 03

Name: Avishek N

Reg No: 192511009

Program:

```
#include <stdio.h>

int recursiveFactorial(int n)
{
    if (n == 0 || n == 1)
        return 1;
    return n * recursiveFactorial(n - 1);
}

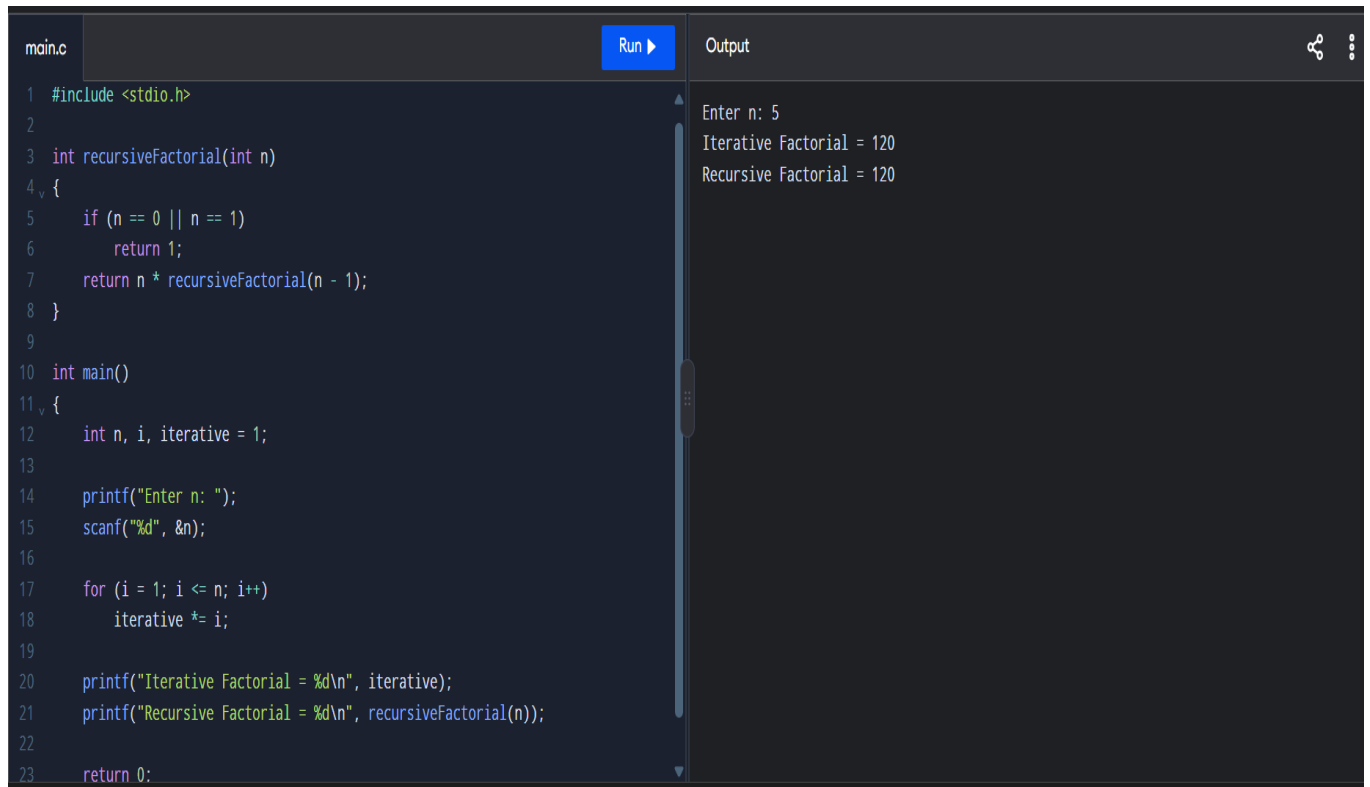
int main()
{
    int n, i, iterative = 1;
    printf("Enter n: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
        iterative *= i;

    printf("Iterative Factorial = %d\n", iterative);
    printf("Recursive Factorial = %d\n", recursiveFactorial(n));

    return 0;
}
```

Output:



The image shows a code editor window with a file named 'main.c'. The code defines a recursive factorial function and a main function that takes input 'n' and calculates both iterative and recursive factorials. The output window shows the program's execution results for 'n = 5'.

```
main.c Run ▶ Output
```

```
1 #include <stdio.h>
2
3 int recursiveFactorial(int n)
4 {
5     if (n == 0 || n == 1)
6         return 1;
7     return n * recursiveFactorial(n - 1);
8 }
9
10 int main()
11 {
12     int n, i, iterative = 1;
13
14     printf("Enter n: ");
15     scanf("%d", &n);
16
17     for (i = 1; i <= n; i++)
18         iterative *= i;
19
20     printf("Iterative Factorial = %d\n", iterative);
21     printf("Recursive Factorial = %d\n", recursiveFactorial(n));
22
23     return 0;
```

Enter n: 5
Iterative Factorial = 120
Recursive Factorial = 120